

Advanced (Habanero)

Open Ended Questions (FRQ)

Q9. A factory produces two types of goods, A and B, with x representing the number of units of A, and y representing the number of units of B. The profit from A is 20 per unit, and from B is 30 per unit. The factory has a budget of 1500 for production. The cost of producing x units of A is $10x$, and the cost of producing y units of B is $20y$. If the factory must make a profit of at least 300, and it must produce at least 10 units of A, write the system of inequalities that model the constraint of this situation.

Q10. A company has 2000 to invest in two investments, A and B, with x representing the amount invested in A, and y representing the amount invested in B. The return on A is $0.05x$, and the return on B is $0.10y$. The company must make a return of at least 150. Additionally, the company can invest at most 1500 in A. Write the system of inequalities that model the constraint of this situation, and determine the minimum amount the company needs to invest in B.

Chef's Tips

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- Tip 1: Ensure students understand the context of each problem.
- Tip 2: Encourage students to identify the key constraints and objectives in each scenario.
- Tip 3: Remind students to check their units and ensure dimensional consistency in their calculations.